RES POLYTECHNIC, BANGALORE-91

CIE Practice Test II

Program	Mechanical Engineering	**Semester - III**
Course Name	Programming Concepts in Mechanical Engineering	**Test II**
Course Code	25ME34I	**Marks - 50**
Name of the Course Coordinator	Vinay Kumar	**Duration - 180 min**

Note: Answer any one full question from the following.

Q. No	Questions	CO	Marks
1	a. Write the algorithm and draw the flowchart to find the maximum of three forces F1, F2 and F3.	CO1	5+5+10+ 15+15 = **50 Marks**
	b. Write a Python program to calculate the output speed and output torque of a gear system for the user-entered input speed, input torque and gear ratio using basic operators.	CO1	
	c. Write a Python program to calculate the factorial of a number using a while loop.	CO1	
	d. Write a Python program that stores the names of students in a list and performs the operations of adding a student, removing a student and sorting the list, displaying the list after each operation.	CO1	
	e. Write a Python class for a bank account that supports deposit, withdrawal and balance inquiry. Display a suitable message when the balance is insufficient.	CO2	
2	a. Write the algorithm and draw the flowchart to generate the multiplication table of 5 using a control loop.	CO1	5+5+10+ 15+15 = **50 Marks**
	b. Write a Python program to accept two numbers from the user and perform the arithmetic operations addition, subtraction, multiplication and division.	CO1	
	c. Write a Python program to find the greatest common divisor (GCD) of two numbers using a function.	CO1	
	d. Write a Python program to read data from a CSV file and count the number of rows. Use try and except to handle the case when the file is not found.	CO2	
	e. Write a Python class to represent a Car with the attributes model, year and color. Include methods to start, stop and display the car details.	CO2	

Signature of the Course Coordinator

Signature of the HOD

Signature of the IQAC Chairman